

Nasir Sims

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EDUCATION

Washington University in St. Louis

St. Louis, MO

Bachelor of Science in Computer Science, Minor in Film & Media Studies

May 2027

Relevant Courses: Operating Systems, Introduction to Artificial Intelligence, Object-Oriented Programming, Parallel & Concurrent Programming, Programming Systems & Languages, Digital Logic & Computer Design, Data Structures & Algorithms

PROJECTS

Bidirectional Camera System | *Raspberry Pi, Bash, Soldering*

Nov. 2024 - Apr. 2025

- Led the redesign of the rocket's onboard camera system, developing a custom Raspberry Pi Zero 2W-based dual-camera architecture to capture forward and aft flight footage for post-flight analysis and airbrake verification
- Streamlined post-flight data recovery by implementing local video encoding and optimized transfer workflows, while collaborating with avionics and electrical subteams to validate power reliability, and successful data capture during launch

Airbrake Control System | *C++, CMake, Eigen*

Jul. 2024 - Apr. 2025

- Developed a feedforward airbrake system using state estimation and Kalman filtering to optimize aerodynamic drag, leveraging embedded software and PID motor control for precise rocket orientation and altitude control
- Extensively tested embedded software systems, including servo control for paddle extension, altitude verification with elevator tests, and data retrieval via serial monitor output

EXPERIENCE

Chief Safety Officer, WURocketry

Sep. 2023 - Present

Avionics Lead (Jan. - Apr. 2025), Avionics Member (Sep. 2023 - Dec. 2024)

St. Louis, MO

- Serve as the primary liaison between WURocketry and NASA safety officials, ensuring compliance with NASA Student Launch safety requirements, policies, and strict documentation standards
- Refactored our legacy camera system, resulting in on-site viewing of camera footage via Raspberry Pi Zero Microcontroller, and a decrease in footage retrieval from hours to seconds by utilizing local scripting
- Built and maintain an avionics software library to standardize sensor methods using abstract class architecture

Drone Pilot Intern

May 2025 - Aug. 2025

Washington University in St. Louis Facilities Planning & Management

St. Louis, MO

- Piloted DJI Matrice 300 RTK and Mavic 3, planning and executing photogrammetry and LiDAR mapping missions to capture high quality aerial imagery for supporting geospatial research
- Executed LiDAR classification and raster analysis in ArcGIS Pro, applying SLAM-supported workflows to refine ground, building, and vegetation classes for accurate change detection, including seasonal foliage variation and post-tornado damage
- Led the end-to-end internal space audit program, combining Insta360 cameras, ArcGIS Pro, and FME to create the geospatial foundation for a campus-scale digital twin and support business data driven decisions

Research Assistant

Aug. 2024 - Apr. 2025

Independent Study, Washington University in St. Louis

St. Louis, MO

- Worked with a multidisciplinary team of researchers, conducting rigorous stress-testing of Autonomous Vehicles in simulated environments to enhance safety, address vulnerabilities, and improved reliability for real-world deployment
- Worked with a ROS 2-based autonomous navigation framework integrating LaneNet, sensor fusion, and an actor-critic reinforcement learning agent with a neural network employing PPO for lane-following and obstacle avoidance in CARLA simulations and a 1:8-scale urban model, achieving stable performance across diverse traffic and environmental conditions
- Customized LaneNet using TensorFlow to enhance lane detection at intersections, significantly improving accuracy

InfoSec GRC Analyst Intern

Apr. 2024 - Present

Washington University in St. Louis

St. Louis, MO

- Deliver timely updates to information security policies to meet compliance standards and audit schedules, ensuring consistent security practices across all university departments
- Lead risk assessments for WashU's 1000+ vendor network, utilizing OneTrust and Black Kite security platforms to streamline compliance, reduce cybersecurity incidents, and adhere to evolving AI-driven industry standards
- Manage and update a glossary of 200+ IT terms to maintain accurate documentation for organizational use

SKILLS

Languages: C++, C, Java, Python, Bash, SML, AVR/RISC-V Assembly, Ruby, Racket, Intermediate Mandarin Chinese

Software: ArcGIS Pro, FME, Esri Drone2Map, GeoSLAM, Pix4D, Office 365, MySQL, Tableau

Libraries: CARLA, PyTorch, TensorFlow, Gymnasium, scikit-learn, BSXlite, RadioHead, Pandas, Matplotlib, NumPy, Eigen